

```
theory Week_1_sol
imports Main
begin
```

King's College London
Department of Informatics
Dr. Mohammad Abdulaziz

Term 1

Computer-Assisted Theorem Proving: 7CCSACTP

Exercise Sheet 1

Important Note: Please download Isabelle from the link provided in the slides and bring a laptop to the large group tutorial. Performing the proofs in Isabelle will be an integral part of those tutorials.

Before you start, please open a new Isabelle/HOL theory file called `Week_1.thy` and add the following to its beginning.

```
theory Week_1
imports Main
begin
```

Exercise 1.1 Trying out Isabelle

In this sheet, you should try out the Isabelle theorem prover. Prove the following theorem, first on pen-and-paper, then formally in Isabelle.

lemma

```
assumes T: T b
  and A: A a  $\wedge$  A b  $\implies$  a = b
  and TA:  $\bigwedge x. T x \implies A x$ 
  and Aab: A a
shows T a
```

proof—

— Style: declarative (Isar *have/show*) with apply-style steps; forward (*OF*); backward (*rule*); *subst*.

```
have 1: A b
  using TA[OF T]
  .
```

```

have  $\mathscr{Q}$ :  $A\ a \wedge A\ b$ 
  apply(rule conjI)
  using  $Aab\ 1$ 
  .

have  $\mathscr{Q}$ :  $a = b$ 
  apply(rule A)
  using  $\mathscr{Q}$ 
  .

show ?thesis
  using  $T$ 
  apply(subst  $\mathscr{Q}$ )
  .
qed

end

```